

Reddit Clone Project Report

Team Report

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1 Team Identification

Name	Student ID
Islam Nashaat Ahmed	23P0077
Ahmad Wail	23P0335
Omar Hany Hussein Mohamed	23P0424
Youssef Ahmed Farouk	23P0054
Malek Wael Mohamed	23P0063

2 GitHub repository:

<https://github.com/yousseffffahmedddd/Reddit>



3 Work Distribution

- **Islam Nashaat Ahmed:** Frontend UI development and component implementation.
- **Ahmad Wail:** Backend development and frontend API integration layer.
- **Youssef Ahmed Farouk:** Backend development and frontend API integration layer.
- **Omar Hany Hussein Mohamed:** System integration and stability checks.
- **Malek Wael Mohamed:** Feature integration and end-to-end testing.

4 Database Schema

Below is the Entity-Relationship (ER) diagram representing the database schema used for this project.

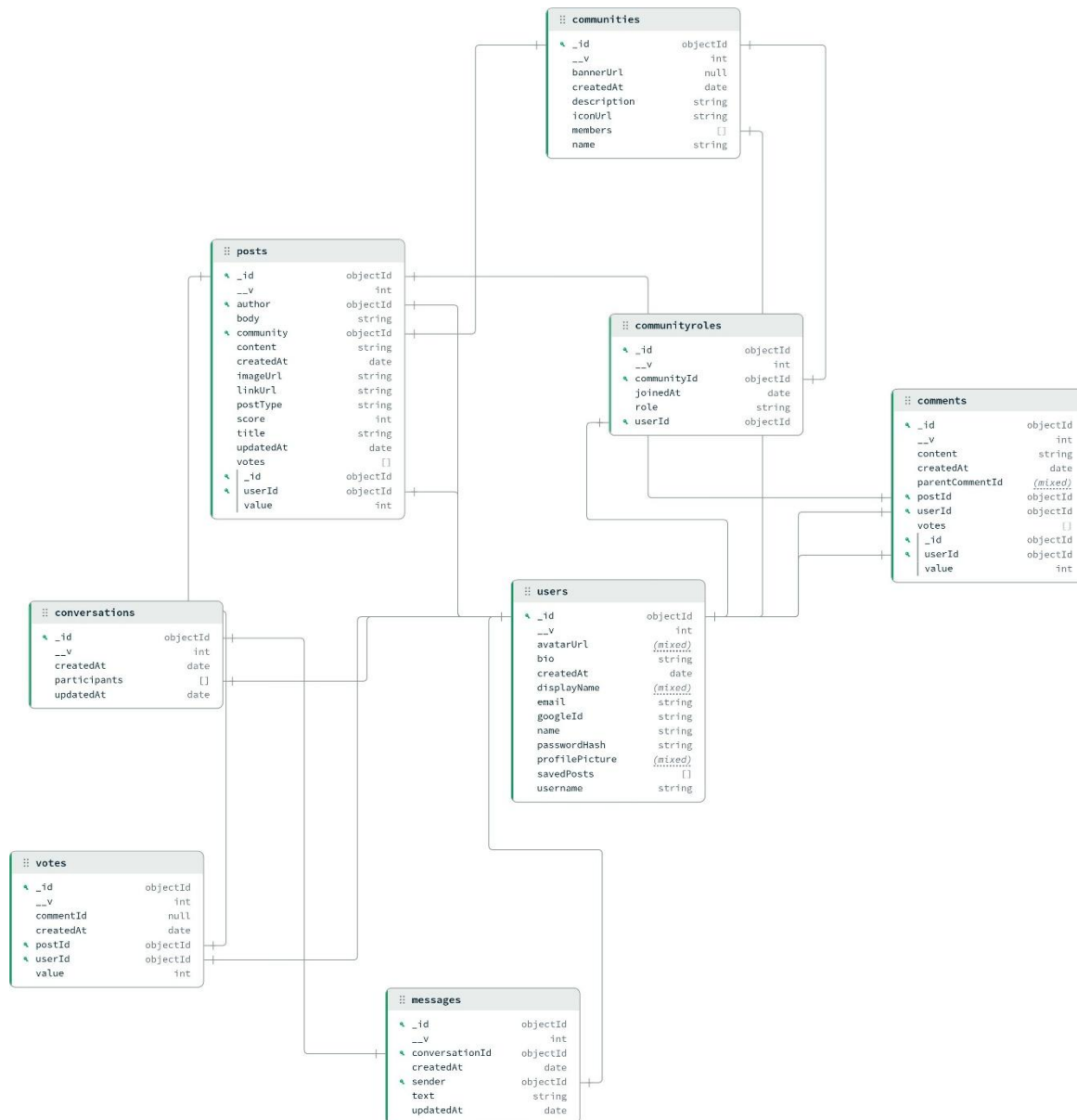


Figure 1 Project Database Schema

5 API Documentation

5.1 Environment Configuration

The API supports multiple environments. Configuration is managed via `http-client.env.json`.

```
1 {  
2   "dev": {  
3     "baseUrl": "http://localhost:3000",  
4     "userId": "YOUR_USER_ID",  
5     "postId": "YOUR_POST_ID",  
6     "communityId": "YOUR_COMMUNITY_ID",  
7     "commentId": "YOUR_COMMENT_ID",  
8     "conversationId": "YOUR_CONVERSATION_ID",  
9     "username": "testuser"  
10  }  
11 }
```

Listing 1: Environment Config Example

5.2 Endpoints Summary

5.2.1 Authentication (/api/auth)

Method	Endpoint	Description
POST	/api/auth/signup	Register a new user
POST	/api/auth/login	Login with email/password
POST	/api/auth/google	Google OAuth login
POST	/api/auth/logout	Logout

5.2.2 Users (/api/users)

Method	Endpoint	Description
GET	/api/users/profile/:username	Get user profile
PUT	/api/users/profile	Update user profile
POST	/api/users/profile/picture	Upload profile picture
DELETE	/api/users/profile/picture	Delete profile picture
POST	/api/users/save-post	Save/unsave a post
GET	/api/users/saved-posts/:userId	Get saved posts

5.2.3 Communities (/apis/Communityapi)

Method	Endpoint	Description
GET	/apis/Communityapi	Get all communities
GET	/apis/Communityapi/search?q=	Search communities
POST	/apis/Communityapi	Create a community
GET	/apis/Communityapi/:communityId	Get community by ID/name
GET	/apis/Communityapi/user/:userId	Get joined communities
GET	/apis/Communityapi/owned/:userId	Get owned communities
POST	/apis/Communityapi/join	Join a community
PUT	/apis/Communityapi/:communityId	Update community
POST	/apis/Communityapi/:communityId/icon	Upload community icon

5.2.4 Posts (/apis/Postapi)

Method	Endpoint	Description
GET	/apis/Postapi	Get all posts
GET	/apis/Postapi/:postId	Get post by ID
GET	/apis/Postapi/popular	Get popular posts
POST	/apis/Postapi	Create a post
POST	/apis/Postapi/upload	Upload post image
PUT	/apis/Postapi/:postId	Update a post
DELETE	/apis/Postapi/:postId	Delete a post
POST	/apis/Postapi/vote	Vote on a post (legacy)

5.2.5 Votes (/api/votes)

Method	Endpoint	Description
POST	/api/votes	Vote on a post

5.2.6 Comments (/api/comments)

Method	Endpoint	Description
GET	/api/comments/post/:postId	Get comments for a post
GET	/api/comments/post/:postId/count	Get comment count
GET	/api/comments/:commentId	Get single comment
GET	/api/comments/user/:userId	Get user's comments
POST	/api/comments	Create a comment
PUT	/api/comments/:commentId	Update a comment
DELETE	/api/comments/:commentId	Delete a comment
POST	/api/comments/:commentId/vote	Vote on a comment

5.2.7 Chat (/api/chat)

Method	Endpoint	Description
GET	/api/chat/users	Get all users
GET	/api/chat/users/search?q=	Search users
POST	/api/chat/conversation	Get/create conversation
GET	/api/chat/conversations/:userId	Get user's conversations
GET	/api/chat/messages/:conversationId	Get messages
POST	/api/chat/message	Send a message

5.2.8 AI Features

Method	Endpoint	Description
POST	/api/chatbot/ask	Ask AI chatbot
POST	/apis/ai/summarize	Summarize a post

5.3 Request/Response Examples

Create a Post

```
1 POST {{baseUrl}}/apis/Postapi
2 Content-Type: application/json
3
4 {
5   "title ": "My Post Title ",
6   "content": "Post content here",
7   "postType": "text",
8   "author ": "userId ",
9   "community": "communityId "
10 }
```

Vote on a Post

```
1 POST {{baseUrl}}/api/votes
2 Content-Type: application/json
3
4 {
5   "postId": "postId",
6   "userId": "userId",
7   "value": 1
8 }
```

Vote values: 1 = Upvote, -1 = Downvote, 0 = Remove vote.

5.4 Real-Time Features (Socket.IO)

The application connects to ws://localhost:3000 for live updates.

Event	Direction	Description
join	Client → Server	Join a conversation room
sendMessage	Client → Server	Send a message
newMessage	Server → Client	Receive a new message
typing	Both	Typing indicator